ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO ROBOTICS

PAPER 2 · PRACTICE B

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Paper 2 style (design-related), built around one case study: an autonomous warehouse delivery robot. Section A: objective questions (30) · Section B: system design and battery runtime, wheel speed, gearing and force calculations, obstacle-avoidance flowchart and Arduino code, IoT reporting, a 3D-printed part and a risk assessment (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Draw diagrams and flowcharts neatly, using a ruler. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 A differential-drive robot changes direction by …

A turning its front wheels.
B changing the speeds of its left and right wheels.
C using a rudder.
D moving its legs.

1.1.2 The motor-driver board often used to control the direction and speed of two DC motors is the …

A L298N. B 7805. C LM35. D HC-SR04.

1.1.3 A gearbox that reduces the speed of a motor …

A has no effect on torque.
B reduces the torque.
C increases the torque.
D changes the supply voltage.

1.1.4 The linear speed of a wheel with diameter d (in m) turning at N revolutions per minute is …

A $d \div (\pi N)$ B $N \div (60\pi d)$ C $\pi d \div N$ D $\pi dN \div 60$

1.1.5 The Arduino function that waits for a pulse on a pin and returns its length is …

A delay() B pulseIn() C millis() D map()

1.1.6 The Arduino function millis() returns …

A the time since the board started, in milliseconds.
B the temperature.
C the distance to an object.
D an analogue reading.

1.1.7 A lightweight messaging protocol often used for IoT devices is …

A FTP. B SMTP. C MQTT. D POP3.

1.1.8 The file format most often used to send a 3D model to slicing software is …

A .mp3 B .docx C .stl D .exe

1.1.9 The 3D-printing filament that is easy to print and made from plant starch is …

A ABS. B PLA. C nylon. D carbon steel.

1.1.10 A lithium-ion battery pack must be protected by a …

A battery management system (BMS). B light-dependent resistor.
C liquid-crystal display. D PIR sensor.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 In a flowchart, a diamond shape shows a decision.

1.2.2 While delay() is running, the microcontroller keeps reading its sensors.

1.2.3 An H-bridge can reverse the direction of a DC motor.

1.2.4 At the same rpm, a robot with larger wheels moves more slowly.

1.2.5 PWM can be used to control the speed of a DC motor.

1.2.6 Connecting two identical batteries in series doubles the capacity in mAh.

1.2.7 The infill percentage affects the strength of a 3D-printed part.

1.2.8 Wi-Fi reaches every corner of a large steel warehouse without any extra access points.

1.2.9 A risk assessment identifies hazards and the measures to control them.

1.2.10 Encoders can be used to keep two wheels turning at the same speed.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Shows the steps and decisions of a program with standard shapes	A	infill
1.3.2	Circuit of four switches that can reverse a motor	B	PWM
1.3.3	Set of gears that lowers speed and raises torque	C	flowchart
1.3.4	Circuit that protects lithium cells from over-charge and over-discharge	D	slicer
1.3.5	Software that cuts a 3D model into printing layers	E	dashboard
1.3.6	How much of the inside of a 3D print is filled	F	H-bridge
1.3.7	Online screen that shows live data from the robot	G	state machine
1.3.8	Program design where the robot is always in one of a set of modes	H	battery management system
1.3.9	Sensor that detects the body heat of a moving person	I	PIR sensor
1.3.10	Switching output whose on-time sets a motor's speed	J	gearbox

(10)

TOTAL SECTION A: [30]

SECTION B

CASE STUDY. A company asks you to design a small autonomous robot that carries parcels of up to 5 kg along the aisles of a warehouse. It must follow the aisles, stop for people and obstacles, report its battery level to the supervisor's computer and run for at least one hour on a charge.

QUESTION 2: SYSTEM DESIGN

2.1 Draw a neat, labelled block diagram of the delivery robot showing the power supply, the controller, at least THREE sensors, the actuators and the communication link. (6)

2.2 Choose between a differential-drive and a car-like (Ackermann) steering system for this robot. Give a reason for your choice. (3)

2.3 Write FOUR clear design specifications (requirements) for the robot. (4)

2.4 The robot uses a 7,4 V, 4 000 mAh battery, of which only 80% may be used. Each of the two drive motors draws 1,2 A, the electronics draw 0,4 A and the sensors 0,2 A. Calculate the running time in hours and minutes, and state whether the robot meets the one-hour requirement. (5)

[18]

QUESTION 3: MOTION CALCULATIONS

3.1 The wheels have a diameter of 80 mm and turn at 120 r/min. Calculate the speed of the robot in m/s. (3)

3.2 Calculate how long the robot takes to travel along a 30 m aisle. (2)

3.3 Each motor turns at 6 000 r/min and drives its wheel through a gearbox. The motor produces a torque of 0,02 N·m.

3.3.1 Calculate the gear ratio needed to give 120 r/min at the wheel. (2)

3.3.2 Calculate the torque at the wheel, ignoring losses. (2)

3.4 Explain how the robot turns left on the spot, and how encoders help it to drive in a straight line even though the two motors are not exactly the same. (4)

3.5 The loaded robot has a mass of 12 kg and must speed up from rest to 0,5 m/s in 2 s. Calculate the acceleration and the force needed. (3)

[16]

QUESTION 4: PROGRAMMING LOGIC

4.1 Draw a flowchart for the following behaviour: the robot drives forward; when an obstacle is closer than 25 cm it stops and waits 3 seconds; if the obstacle is still there it turns right by 90° and carries on, otherwise it simply carries on. (8)

4.2 Study the part of the robot's Arduino program below.

```
const int trigPin = 7;
const int echoPin = 6;

long readCm() {
  digitalWrite(trigPin, LOW);  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH); delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  long duration = pulseIn(echoPin, HIGH);
  return duration * 0.0343 / 2;
}

void loop() {
  if (readCm() < 25) {
    stopAndCheck();
  } else {
    forward();
  }
}
```

4.2.1 What does pulseIn(echoPin, HIGH) measure? (1)

4.2.2 Why is the result divided by 2? (1)

4.2.3 What does the number 0,0343 represent? (1)

4.2.4 Write the function stopAndCheck(). It must stop the motors, wait 3 seconds, measure again and turn right if the obstacle is still closer than 25 cm. Assume that the functions stopMotors(), turnRight90() and readCm() already exist. (6)

4.3 Explain why many robot programs use millis() and a "state machine" instead of long delay() calls. (3)

[20]

QUESTION 5: IoT, 3D PRINTING AND SAFETY

5.1 The robot must send its battery level and position to a dashboard on the supervisor's computer. Name FOUR parts of this IoT system and ONE suitable communication protocol. (4)

5.2 A bracket to hold the ultrasonic sensor will be 3D printed. Describe, in order, FIVE steps from the idea to the finished bracket. (5)

5.3 Choose PLA or PETG for the bracket and give a reason that suits a warehouse. (2)

5.4 Name TWO hazards of this robot to people in the warehouse, and ONE control measure for EACH. (4)

5.5 Where must the emergency-stop button be placed on the robot? (1)

[16]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER